

Rebated property tax: overnight results

Morning review • 9 September 2026

Fixed simultaneous fertility-nest calibration. Annual property tax rises from 1% to 2%; each path rebates its own revenue equally to households.

Start here

Both tax paths completed all eleven dates, 2023–2063. The selected calibration remains fixed at loss 23.791955; the full fit and every estimated parameter follow. Numerical completion does not resolve the economic validation issues below.

The initial fertility gain is +0.509%; by 2063 the birth flow per household changes +1.050%. Young households occupy less space: -4.282% initially and -4.497% in 2063.

The mechanism is clear at impact: lower purchase prices ease the down payment, but the ongoing cost per room rises. The rebate supports births; it does not undo the reduction in housing space.

Immediate discussion: (1) the first-birth housing response remains too small; (2) the policy raises ongoing housing costs despite cheaper assets; (3) the future household-entry rule and almost-universal late-life ownership remain unresolved. These deserve attention before a final presentation claim.

Outcome	2023 impact	2063
Births per household	+0.509 %	+1.050 %
Total births	+0.509 %	+1.251 %
Young ownership	+1.236 pp	+2.154 pp
Young occupied rooms	-4.282 %	-4.497 %
Young-parent ownership	+0.106 pp	+1.746 pp
Young-parent occupied rooms	-3.477 %	-4.184 %
Housing asset price	-4.802 %	-7.591 %

Young = unchanged model age nodes 26, 30 and 34. Young parents have dependent children. These are group averages, with composition effects. Percentage changes and percentage-point changes are distinct.

Complete calibration fit

Moment and units	Target	Model	Gap	Weight	Loss contribution
Completed fertility, cohort centered at age 42 [children]	1.918	1.92871	0.0107128	1425.74	0.163623
Childlessness [share]	0.188	0.18905	0.00105041	17180.7	0.0189567
Mean age at first birth [years]	26.0446	26.2217	0.17712	44.4444	1.39429
First births at age 30 or later [share]	0.260327	0.235629	-0.0246982	10000	6.10002
Housing response to first birth [rooms]	0.720246	0.443859	-0.276388	137.565	10.5086
Parent rooms: 3+ minus 1-2 children, ages 30-55 [rooms]	0.3677	0.383479	0.0157791	2958.51	0.736615
Parent/nonparent ownership gap, ages 30-55 [share]	0.167662	0.161061	-0.00660021	14229.6	0.61988
Ownership, ages 30-55 [share]	0.575472	0.573005	-0.00246662	1207.85	0.00734878
Mean occupied housing, ages 18-85 [rooms]	5.77997	6.21603	0.436061	11.9732	2.27669
Wealth / annual gross labor earnings [ratio]	6.8731	7.00422	0.131121	6.28767	0.108103
Annual bequest flow / aggregate wealth [ratio]	0.0088	0.00854656	-0.000253437	5.16529e+06	0.331769
Wealth/income p90 / p50, ages 76-84 [ratio]	3.44811	3.28443	-0.163681	56.9598	1.52603

All 12 targets and their weights are unchanged. Loss is the sum of weight times squared gap: 23.791955. Completed fertility is a cohort stock; it is distinct from the birth flows reported for the policy paths.

The main misses remain first-birth housing (0.4439 versus 0.7202 rooms) and first births at age 30 or later (0.2356 versus 0.2603).

The calibration table uses the selected calibration environment. The rebated 1% tax reference is a separate policy equilibrium at the same estimated parameters and inherited population; its levels need not equal the calibration moments.

Parameters and restrictions

Parameter	Estimate	Bounds / restriction	Status	Near bound
Annual discount factor	0.995277	[0.94, 0.9995]	Estimated	No
First-birth taste scale	2.16817	[0.02, 50]	Estimated	No
Later-birth taste scale	1.77077	[0.02, 50]	Estimated	No
Owner housing-service premium	1.05373	[0.1, 5]	Estimated	No
Old-state housing normalization	13.716	[0.2, 80]	Estimated	No
Bequest utility weight	0.57035	[0, 8]	Estimated	No
Bequest wealth shift	0.103724	[0.02, 16]	Estimated	Yes
Per-child room floor	0.247085	[0.1, 1.8]	Estimated	No
First-birth utility cost	4.61973	[0, 8]	Estimated	No
First-child room-floor jump	0.469765	[0, 0.5]	Estimated	No
Child-value change, 2007-2023	-0.321388	[-1.5, 0.2]	Estimated	No
Child value in 2007	0.289388	Normalized to old completed fertility of 2.1	Fixed / normalized	No
Child value in 2023	-0.0320002	2007 value plus estimated change	Fixed / normalized	No
Within-nest housing taste scale	0.005	Externally fixed at 0.005	Fixed / normalized	No
Dated housing-supply elasticity	0.63	Externally fixed at 0.63	Fixed / normalized	No

All 11 estimated coordinates retain their original bounds. The first-child room-floor jump is 0.469765 against its 0.5 upper bound; 6.05% of the search span remains. The generic near-bound flag applies to the bequest wealth shift.

The dated supply rule uses elasticity 0.63 without reanchoring for the reform. Old-state construction and the subsequent dated normalization are distinct; this report does not resolve their economic interpretation.

The complete dated policy comparison

Every row compares the 2% tax path with the 1% tax path at the same date. Both return all property-tax revenue through an equal rebate per household. Birth flows include the maintained adjustment for top-coded parity.

Year	Births/HH %	Total births %	HH mass %	Young-parent own. pp	Young-parent rooms %
2023	+0.509	+0.509	+0.000	+0.106	-3.477
2027	+0.608	+0.608	+0.000	+0.064	-4.129
2031	+0.708	+0.708	+0.000	+0.390	-4.443
2035	+0.792	+0.792	+0.000	+1.143	-4.473
2039	+0.859	+0.859	+0.000	+1.294	-4.285
2043	+0.927	+0.950	+0.023	+0.668	-4.364
2047	+0.944	+0.994	+0.050	+0.896	-4.334
2051	+0.891	+0.974	+0.082	+1.560	-4.230
2055	+0.931	+1.049	+0.116	+1.509	-4.316
2059	+0.981	+1.137	+0.155	+1.533	-4.287
2063	+1.050	+1.251	+0.200	+1.746	-4.184

Household mass is identical across paths before 2043. The first policy-induced births enter the household population in 2043 under the maintained twenty-year entry lag.

Why ownership can rise while housing space falls

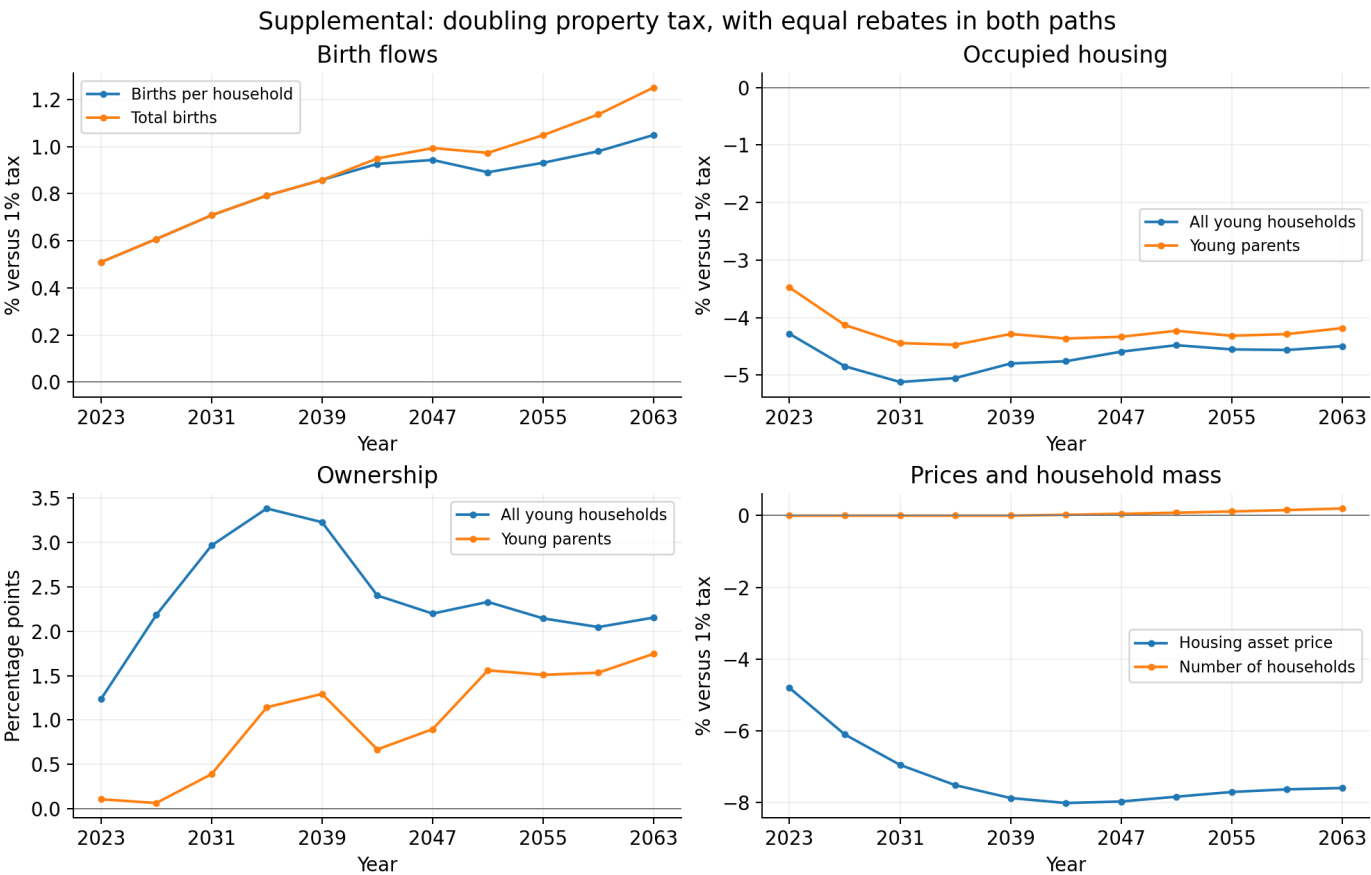
At impact the asset price falls 4.802%, but the four-year rent/user cost per room rises from 0.12106 to 0.14306: +18.178%. The recurring tax more than offsets capitalization in the flow cost. A cheaper purchase therefore does not imply cheaper housing services.

An exact eight-cell decomposition averages each channel over all orderings of changes in the tax rate, asset price and equal rebate. Mixed cells are conditional household responses, not separate equilibria. The components add to the verified impact effect; they are not long-run decompositions.

Outcome	Tax	Price	Rebate	Net
Births/HH (%)	-1.389	+0.305	+1.593	+0.509
Rooms/HH (%)	-6.745	+1.567	+2.124	-3.054
Young ownership (pp)	-0.874	+1.389	+0.721	+1.236
Young rooms (%)	-10.843	+2.883	+3.678	-4.282

Policy effects through 2063

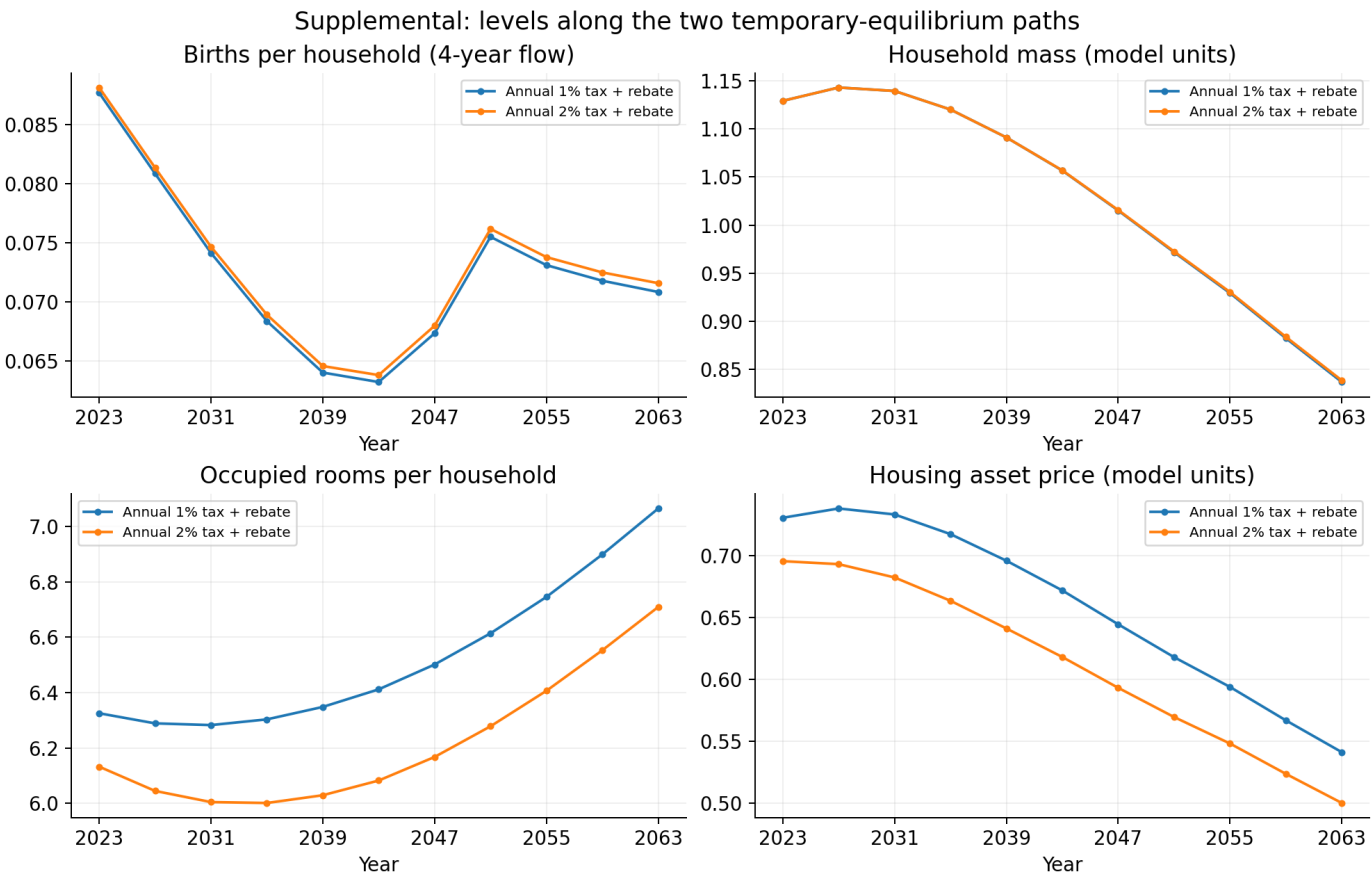
These supplemental summaries complement the unchanged standard diagnostic set. They use the verified comparison table; no additional model is fitted.



Supplemental - Effects relative to the contemporaneous 1% tax path, with equal rebates in both paths. Lines join four-year observations.

Levels along each path

The reference path also changes as the inherited age distribution advances and new household cohorts enter. A percentage difference between policies should not be read as a population forecast.



Supplemental - Model household units and birth flows. The same selected 2023 preference parameters are held fixed throughout.

What is verified, and what remains open

Verified: unchanged scientific source and target fingerprints; common inherited population and entry queues; reproduction of both impact equilibria and both two-date smoke paths; market clearing, balanced rebate ledgers, mass and queue accounting, feasibility, choice probabilities and occupied-state value monotonicity at each date. Full saved checkpoints remain on Torch.

Independent verification covers 26 dated packets, 324 collected artifact hashes and 1,639 checks. All eleven effect rows recompute exactly. Maximum market residual is $1.50843e-5$ against a $2e-4$ gate; maximum fiscal imbalance is $2.42821e-5$ against a $2.5e-5$ gate; mass-accounting error is $1.11022e-15$. Occupied budget-violation mass is zero. All 374 graph files and their checkpoint provenance pass separate hash checks.

Reporting repairs: ownership-policy graphs now sum all five owned sizes and weight conception outcomes; first-birth hazards use the pre-choice population at risk. The corrections reproduce the native transition operator. Fifteen income states have distinct colors; housing and ownership rise with permanent income in independently reconstructed group totals.

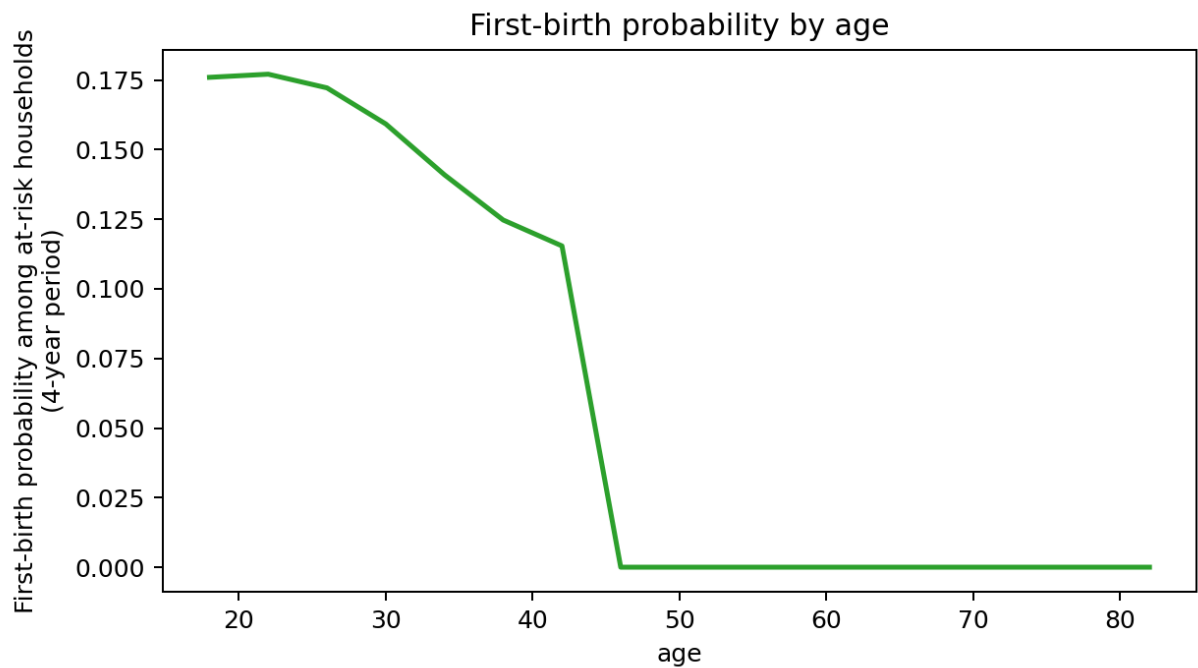
Still open: the first-birth housing response is 0.4439 against 0.7202 rooms. Late-life ownership is close to 100%. Young ownership is 33.1957% in the selected calibration versus an ACS 34.1166%, but exact annual-age alignment remains unresolved. A saved-state check traces ownership-probability dips to renting overtaking four-room ownership before six-room ownership becomes attractive. All inspected origin states have zero population mass. Sensitivity to the numerical grid remains untested; this is neither a demonstrated solver failure nor a proof of grid accuracy.

The forward closure is a closed-national household-unit diagnostic: no outside entry, full retention, births divided by 2.1 for household formation, and a twenty-year lag. Household headship and the handoff from historical empirical age weights to endogenous entrants remain unresolved. It is a sequence of temporary equilibria, with each date's prices perceived as permanent; it is not a perfect-foresight path, resident-population forecast or welfare calculation.

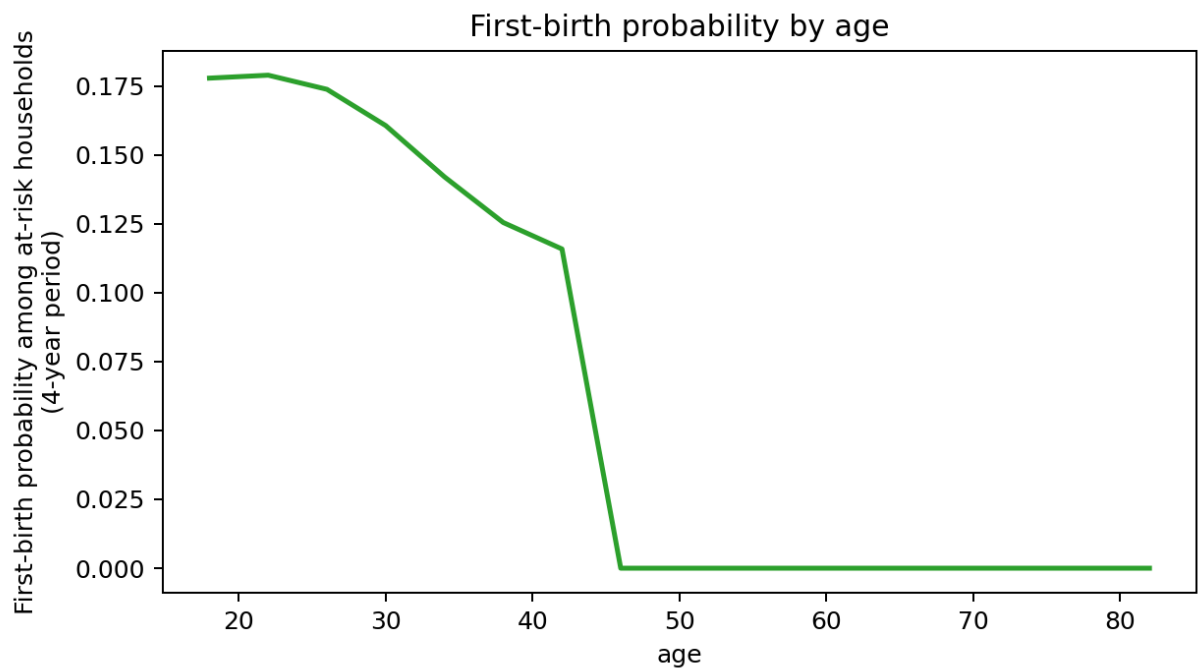
No calibration, weights, target definitions, policy closure, numerical tolerance or production benchmark was changed during this monitoring run. The standard appendix shows both 2063 endpoints; the companion folder contains all 17 standard graphs for every date and both policies (374 images), plus audit receipts.

Standard diagnostic 1: Fertility by age

Both policy paths at 2063. All other dates are in the companion graph set. First births divided by the pre-choice mass at risk, per four-year period.



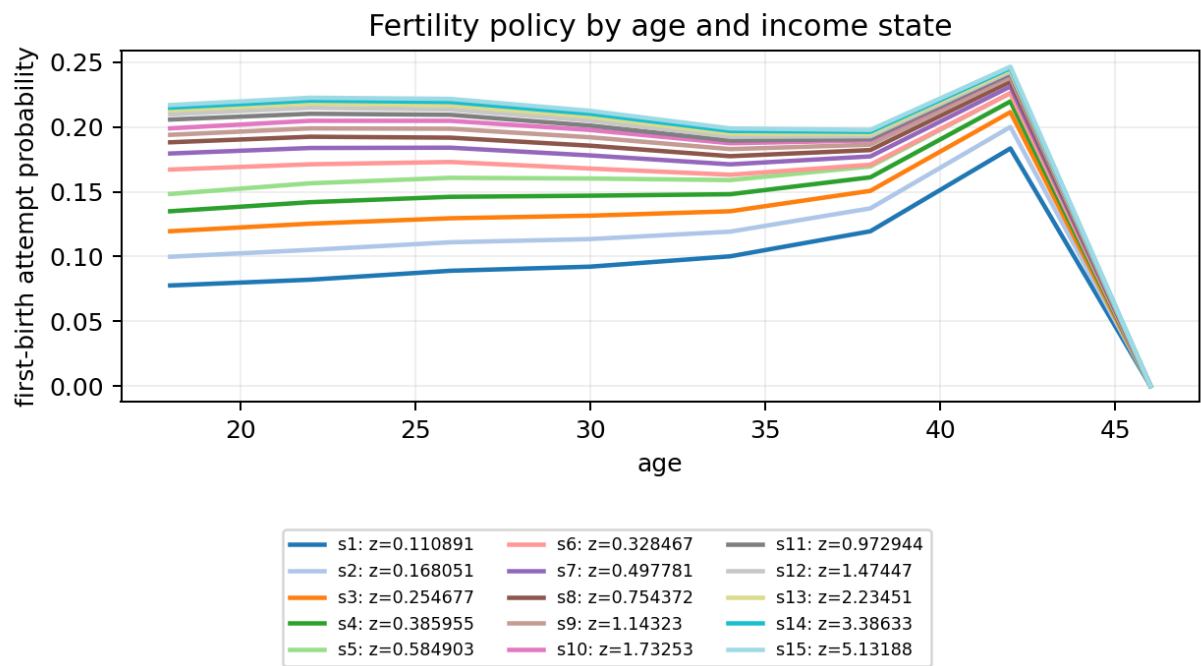
Annual 1% tax + equal rebate • 2063



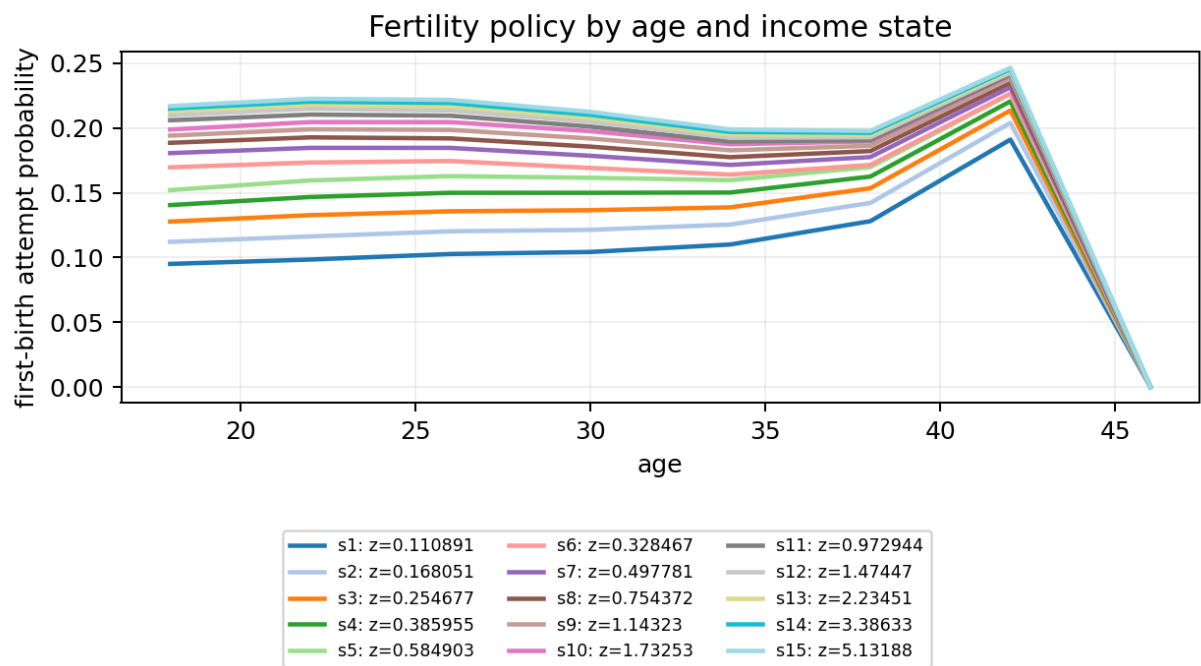
Annual 2% tax + equal rebate • 2063

Standard diagnostic 2: Fertility policy by age income state

Both policy paths at 2063. All other dates are in the companion graph set. Attempt probabilities differ from realized birth probabilities because conception is uncertain.



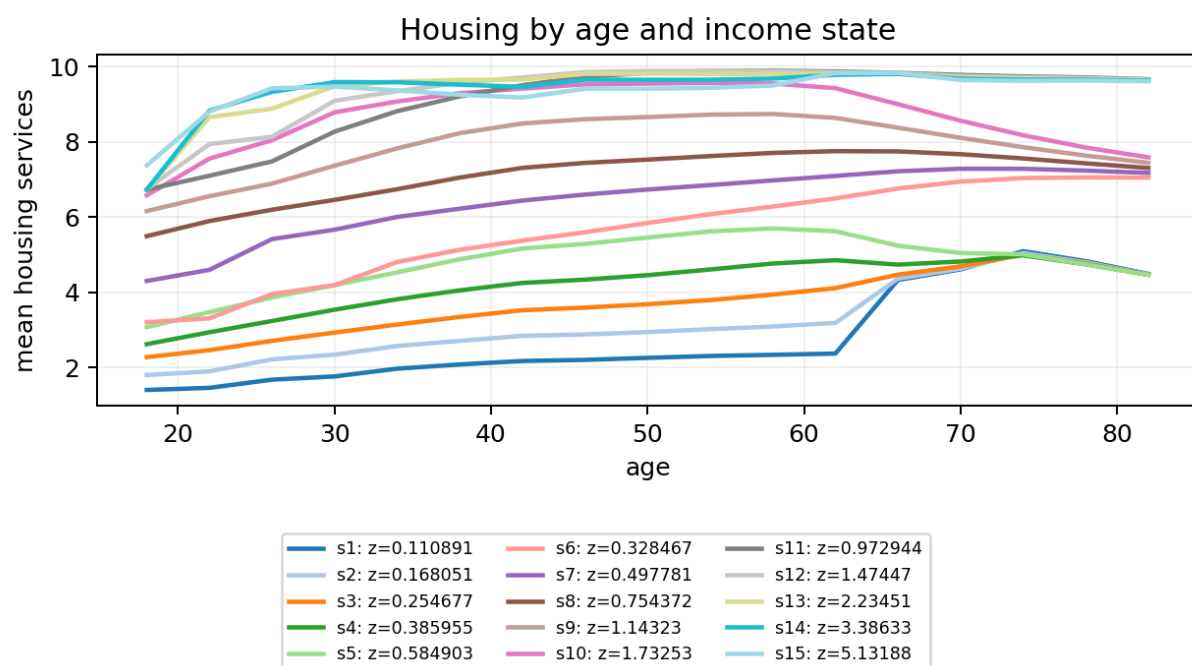
Annual 1% tax + equal rebate • 2063



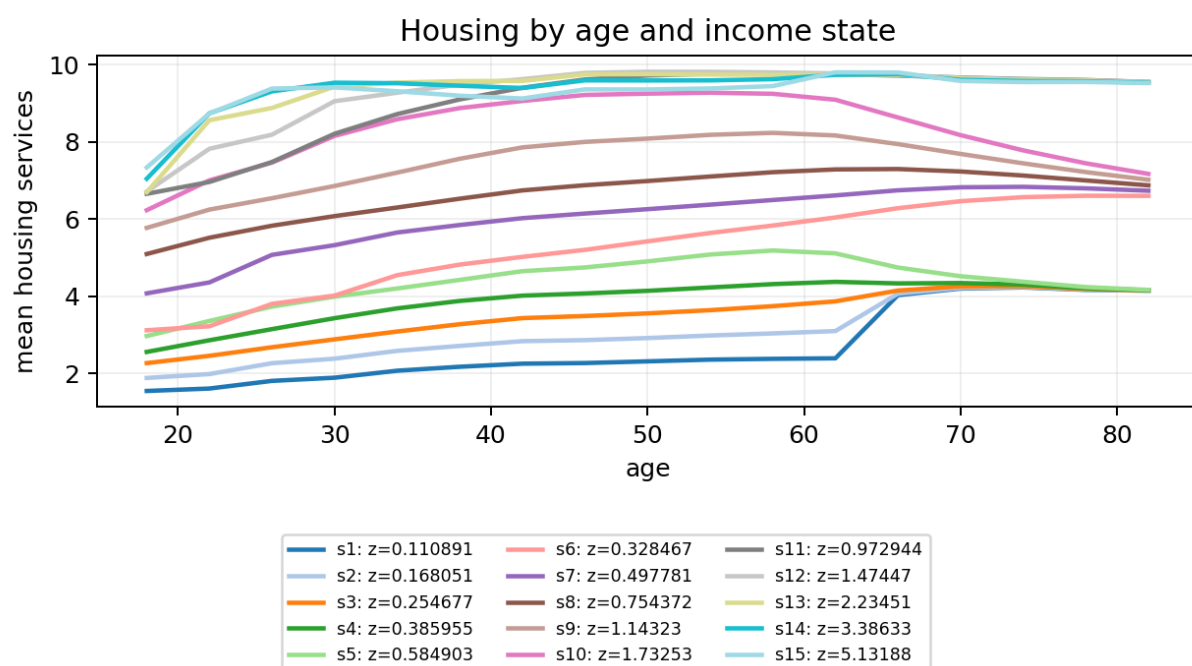
Annual 2% tax + equal rebate • 2063

Standard diagnostic 3: Housing by age income state

Both policy paths at 2063. All other dates are in the companion graph set.



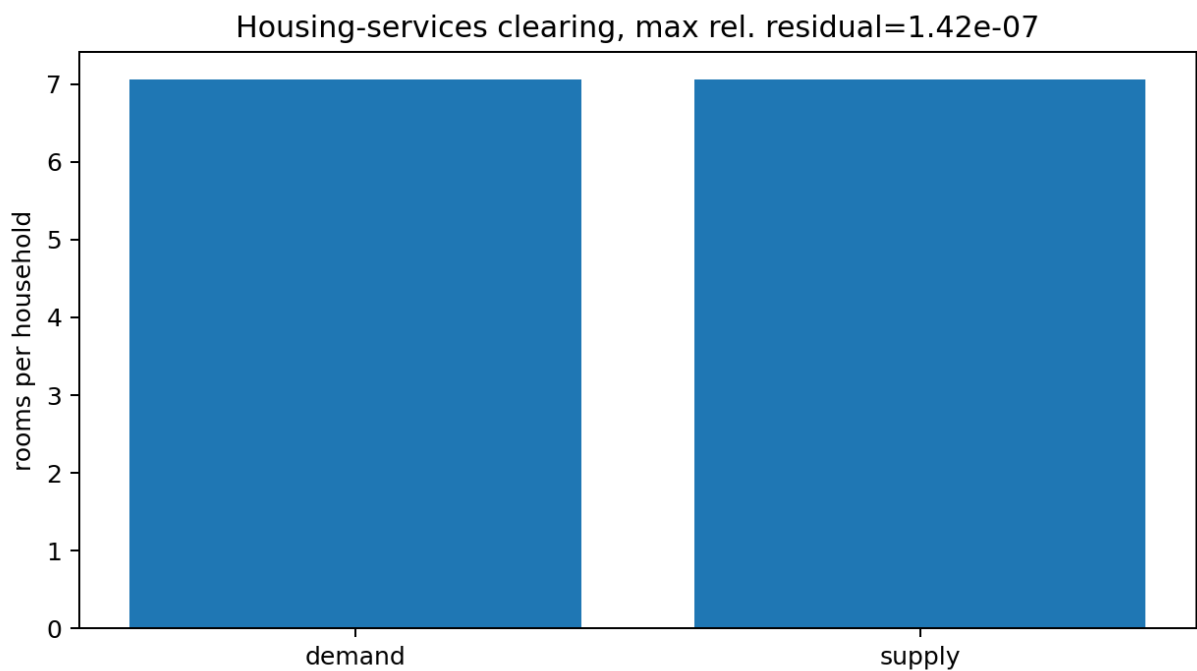
Annual 1% tax + equal rebate • 2063



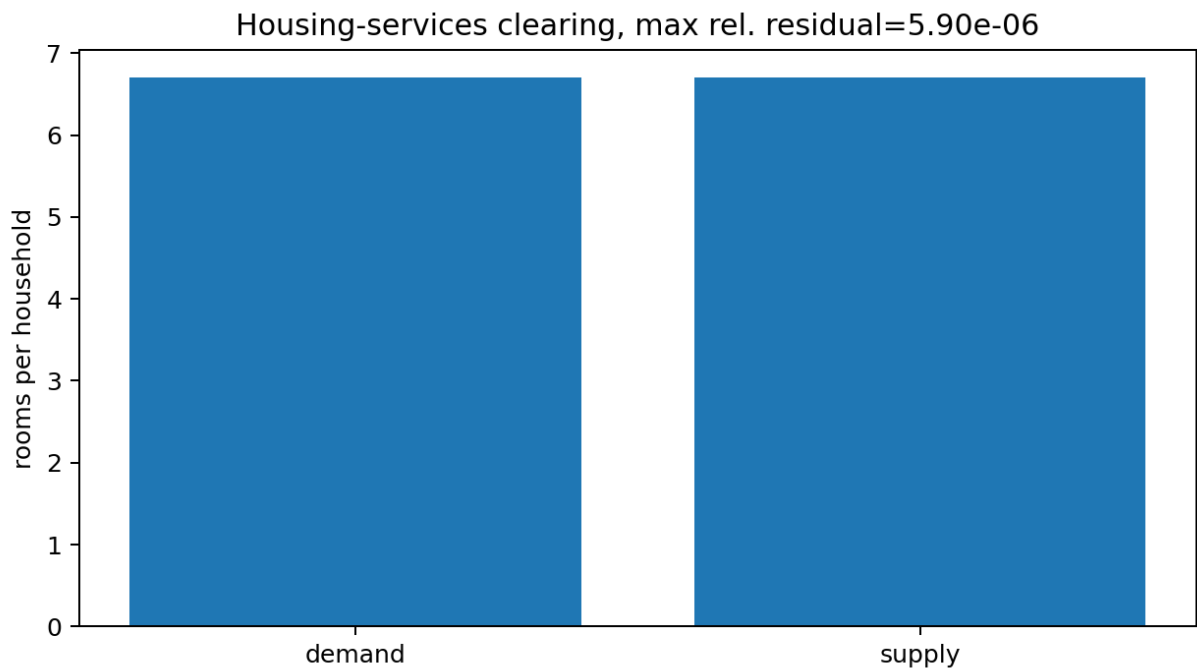
Annual 2% tax + equal rebate • 2063

Standard diagnostic 4: Housing market

Both policy paths at 2063. All other dates are in the companion graph set.



Annual 1% tax + equal rebate • 2063



Annual 2% tax + equal rebate • 2063

Standard diagnostic 5: Housing prices

Both policy paths at 2063. All other dates are in the companion graph set.



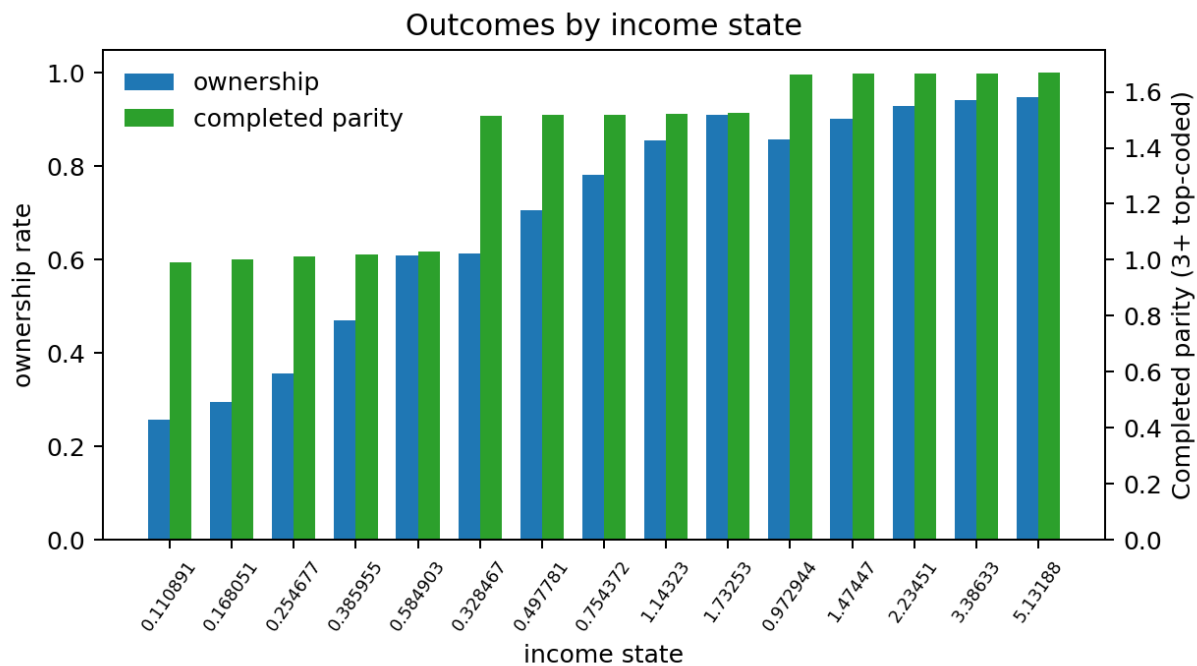
Annual 1% tax + equal rebate • 2063



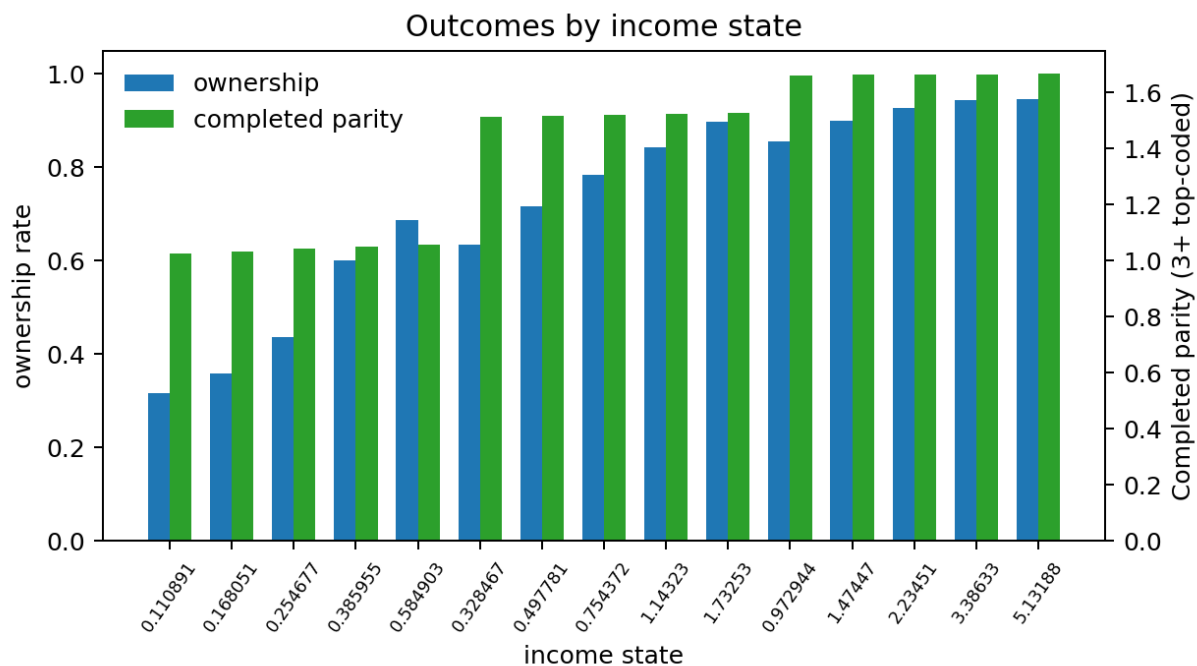
Annual 2% tax + equal rebate • 2063

Standard diagnostic 6: Income state outcomes

Both policy paths at 2063. All other dates are in the companion graph set. Income states retain their permanent-group ordering; the horizontal values are not globally sorted. Parity is top-coded at 3+.



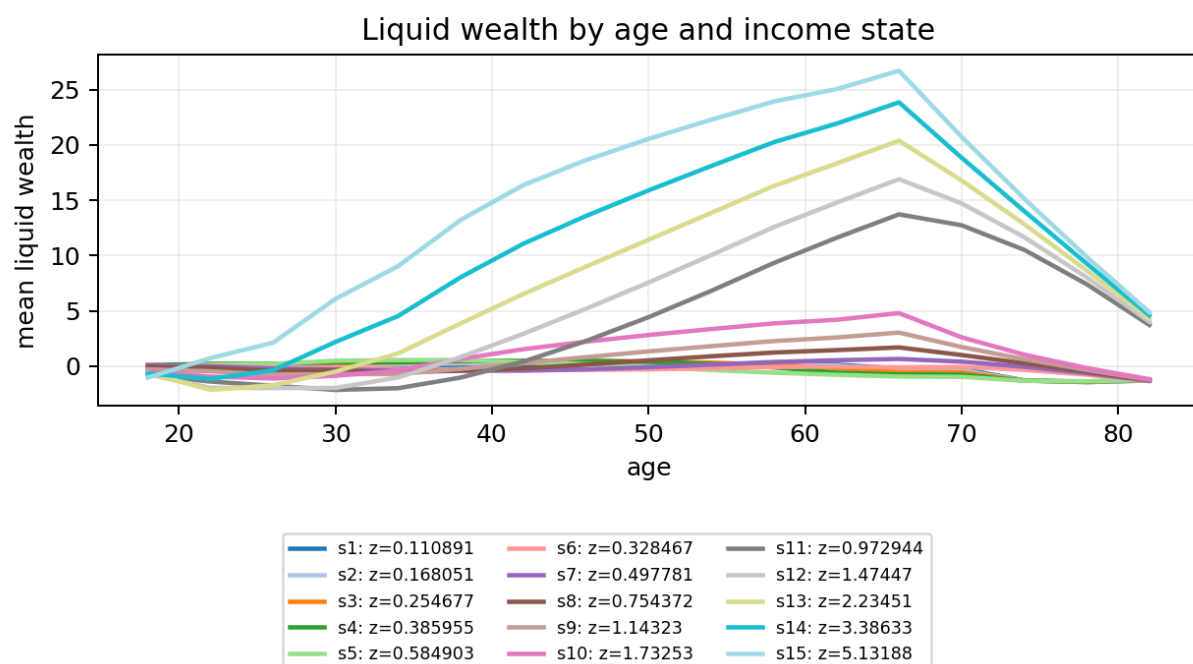
Annual 1% tax + equal rebate • 2063



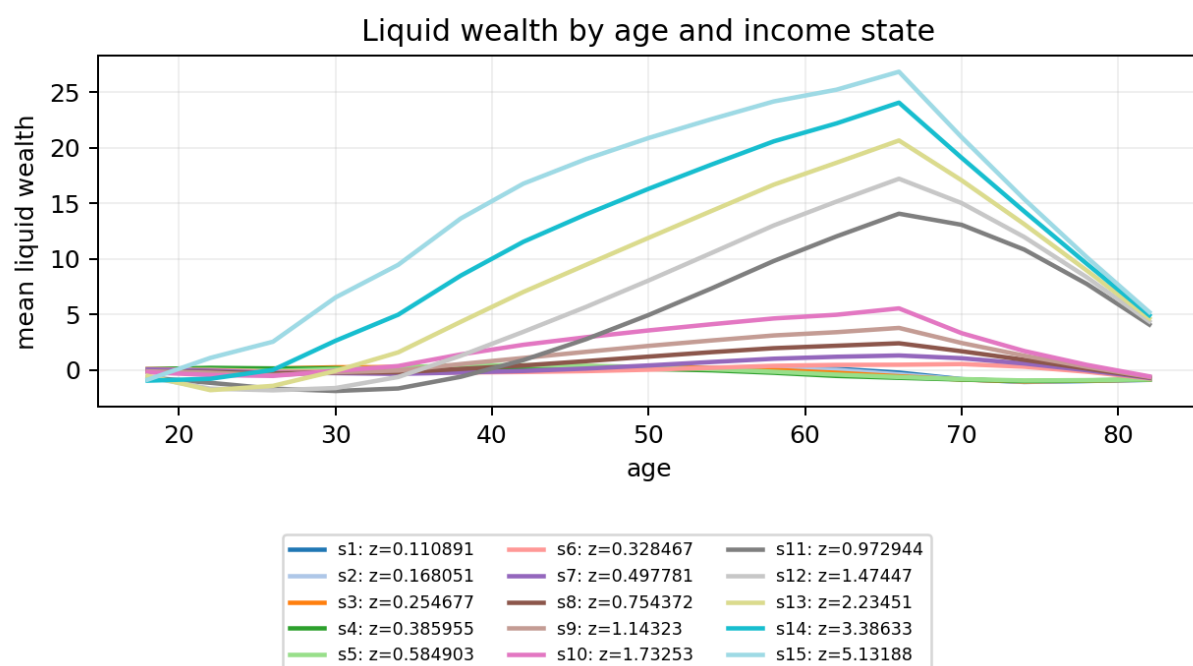
Annual 2% tax + equal rebate • 2063

Standard diagnostic 7: Liquid wealth by age income state

Both policy paths at 2063. All other dates are in the companion graph set.



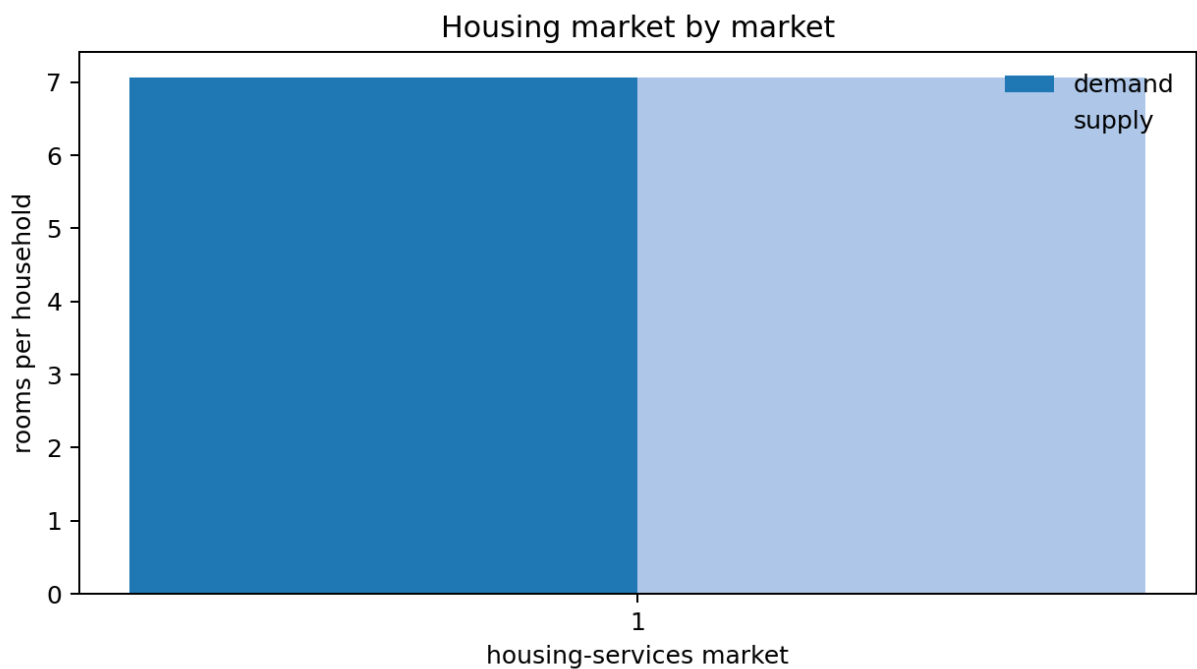
Annual 1% tax + equal rebate • 2063



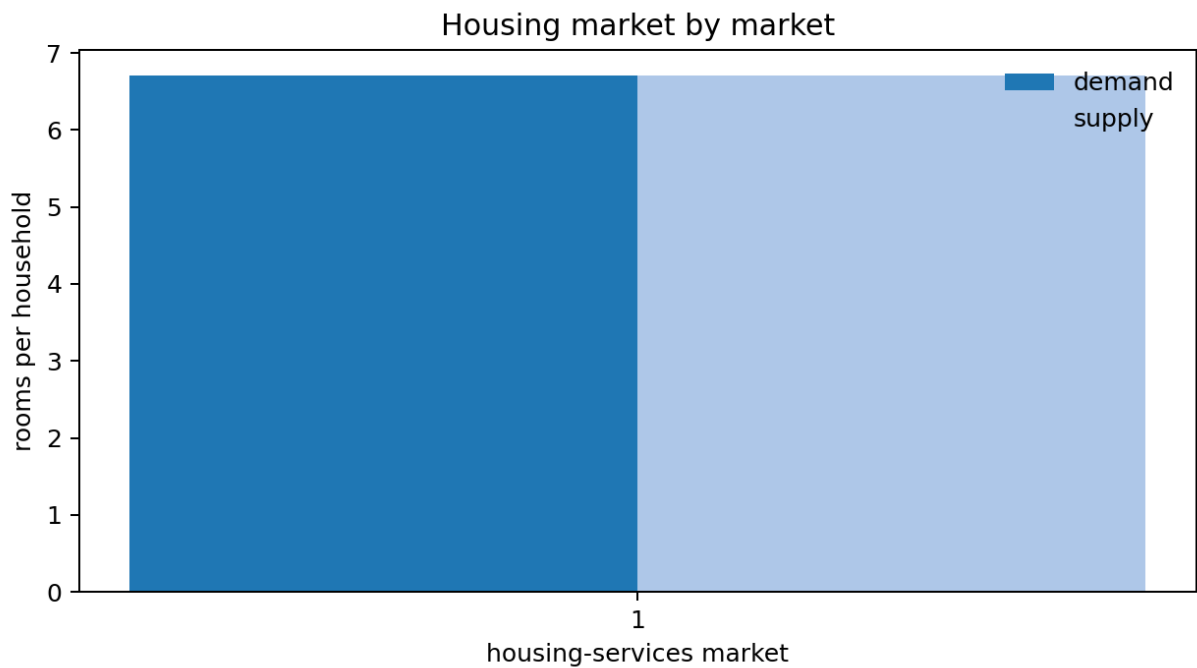
Annual 2% tax + equal rebate • 2063

Standard diagnostic 8: Market clearing by market

Both policy paths at 2063. All other dates are in the companion graph set.



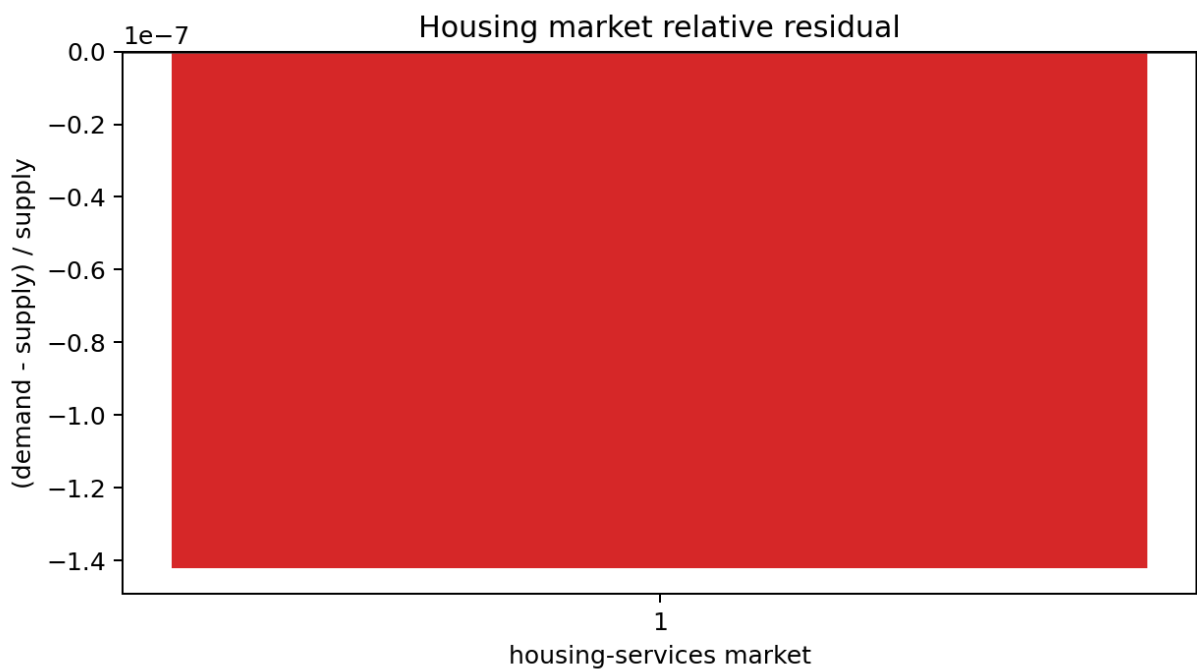
Annual 1% tax + equal rebate • 2063



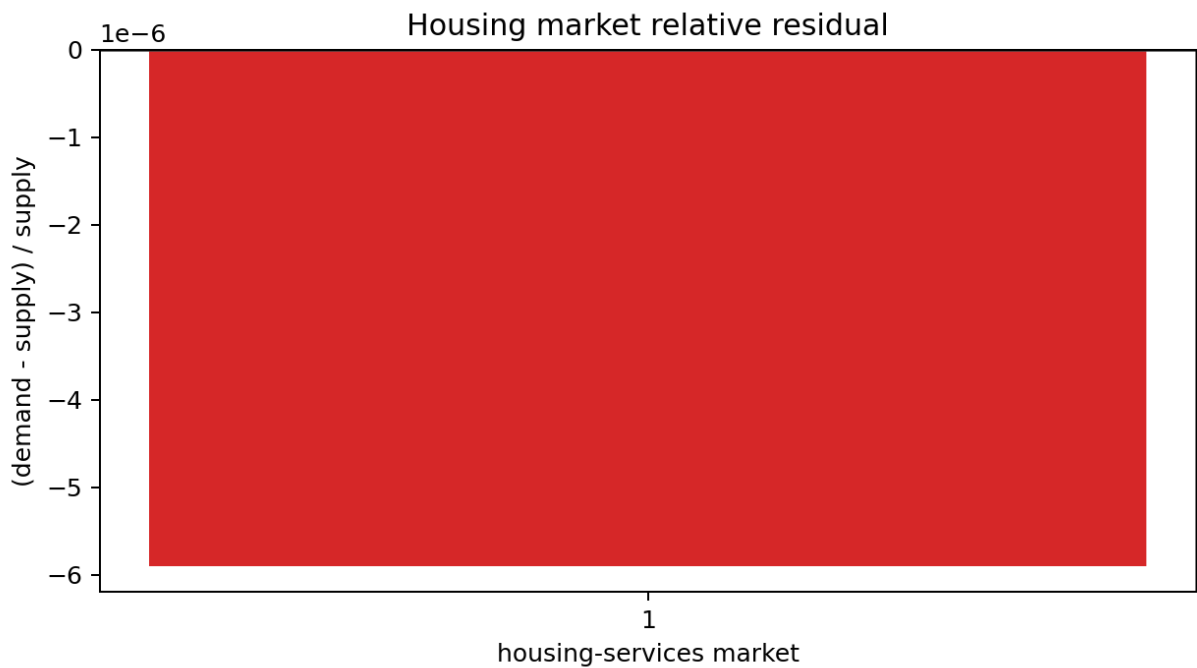
Annual 2% tax + equal rebate • 2063

Standard diagnostic 9: Market clearing residuals

Both policy paths at 2063. All other dates are in the companion graph set.



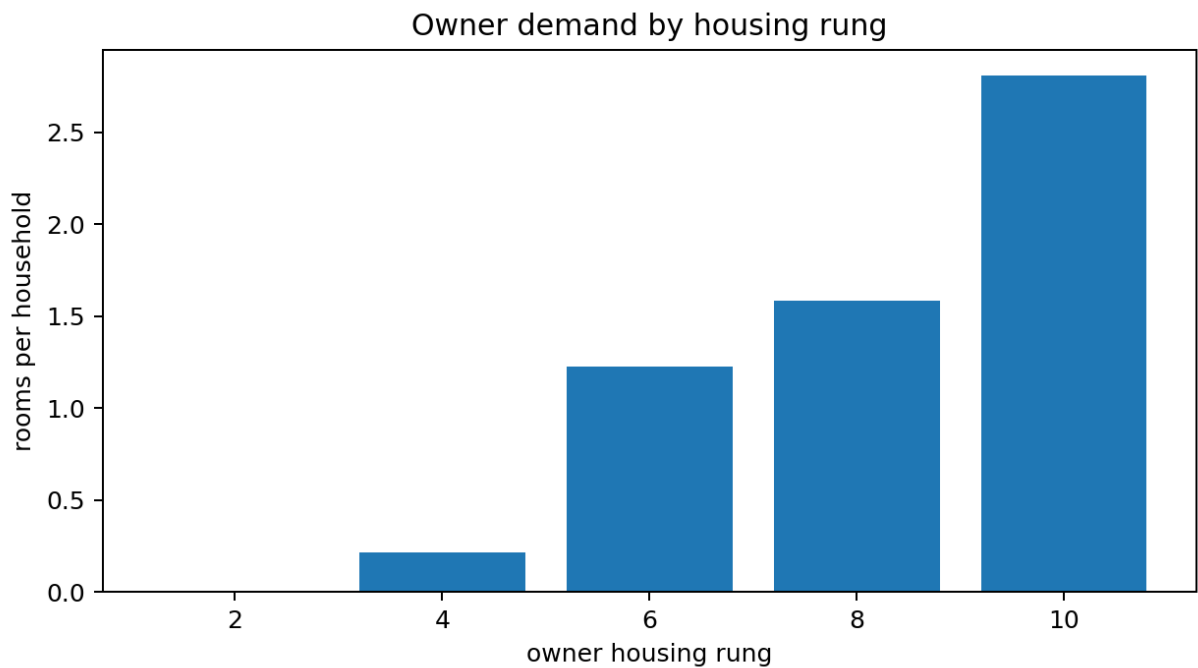
Annual 1% tax + equal rebate • 2063



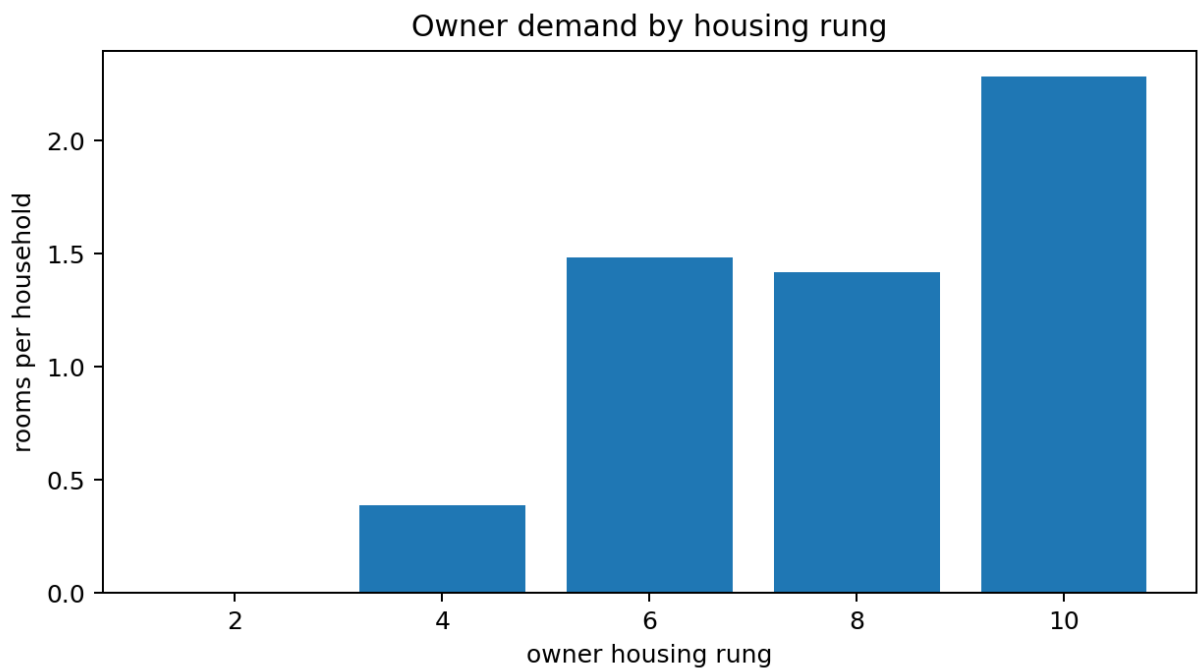
Annual 2% tax + equal rebate • 2063

Standard diagnostic 10: Owner rungs

Both policy paths at 2063. All other dates are in the companion graph set. Bars are contributions to overall rooms per household, not conditional mean rooms within each tenure or housing size.



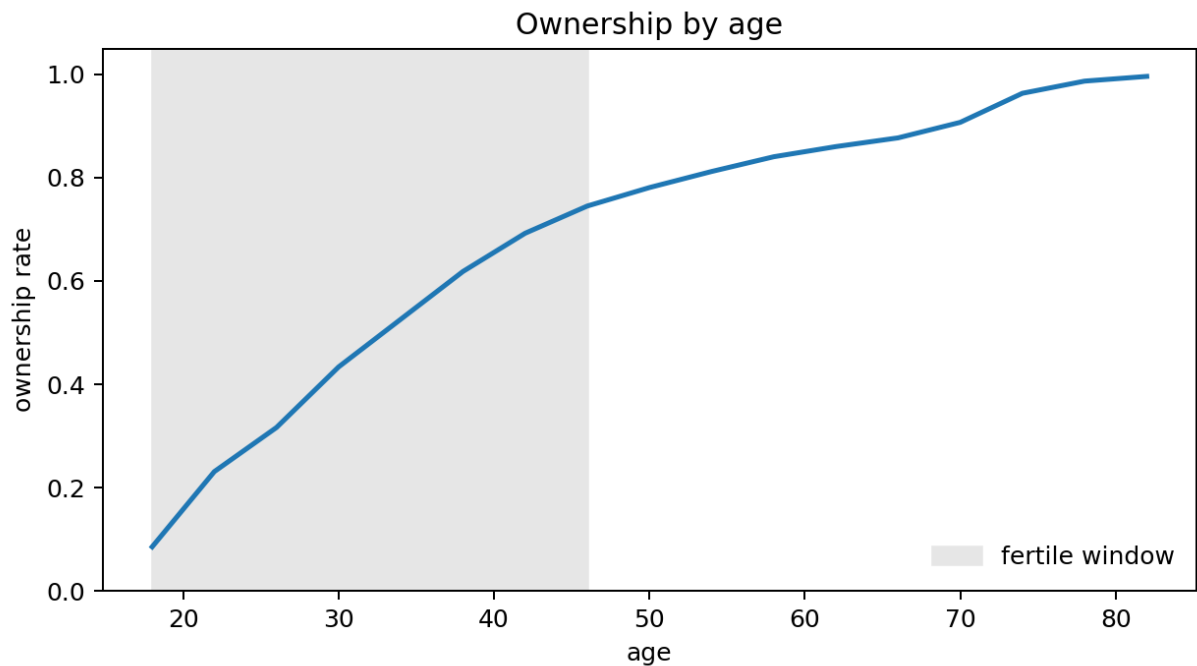
Annual 1% tax + equal rebate • 2063



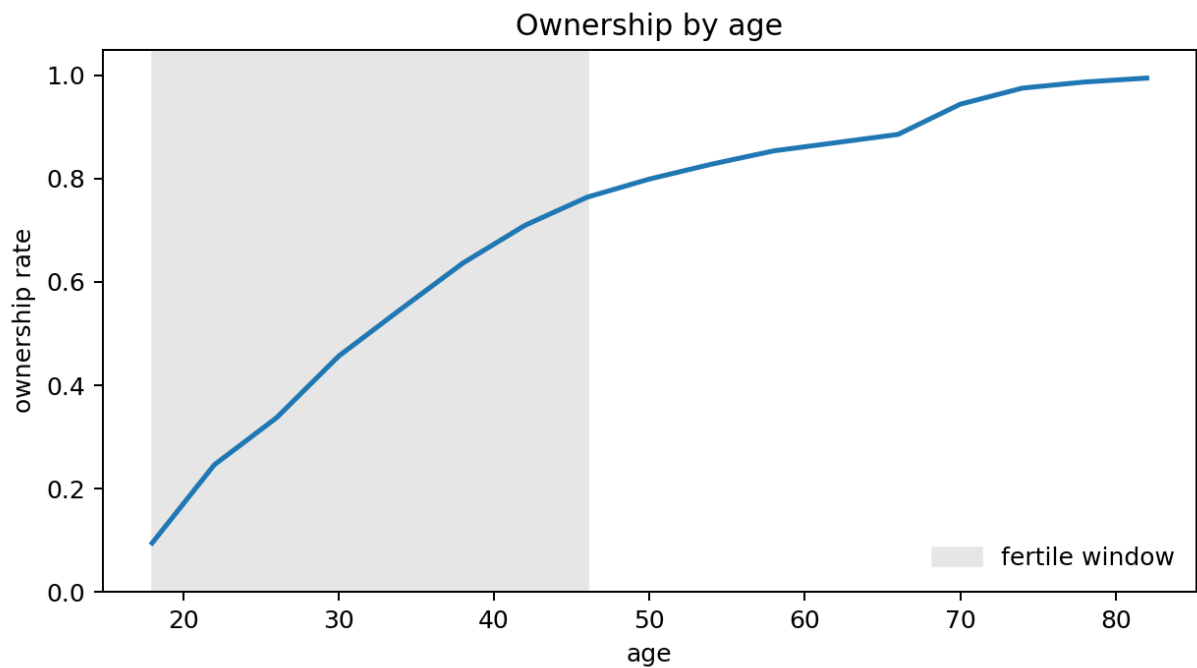
Annual 2% tax + equal rebate • 2063

Standard diagnostic 11: Ownership by age

Both policy paths at 2063. All other dates are in the companion graph set.



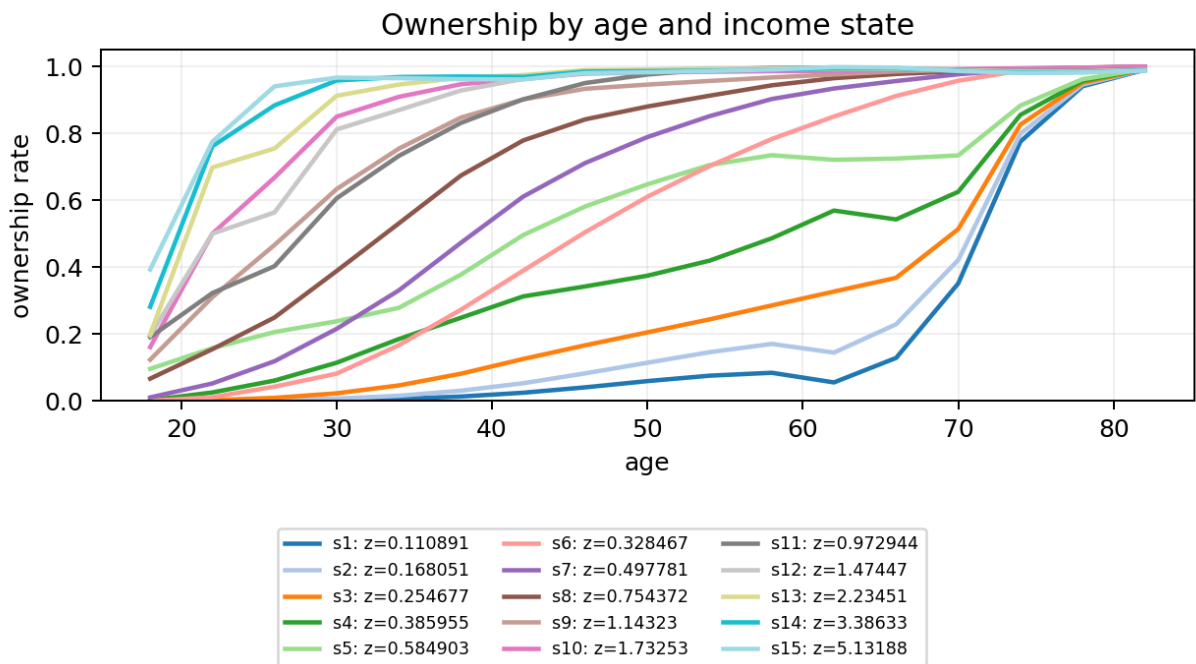
Annual 1% tax + equal rebate • 2063



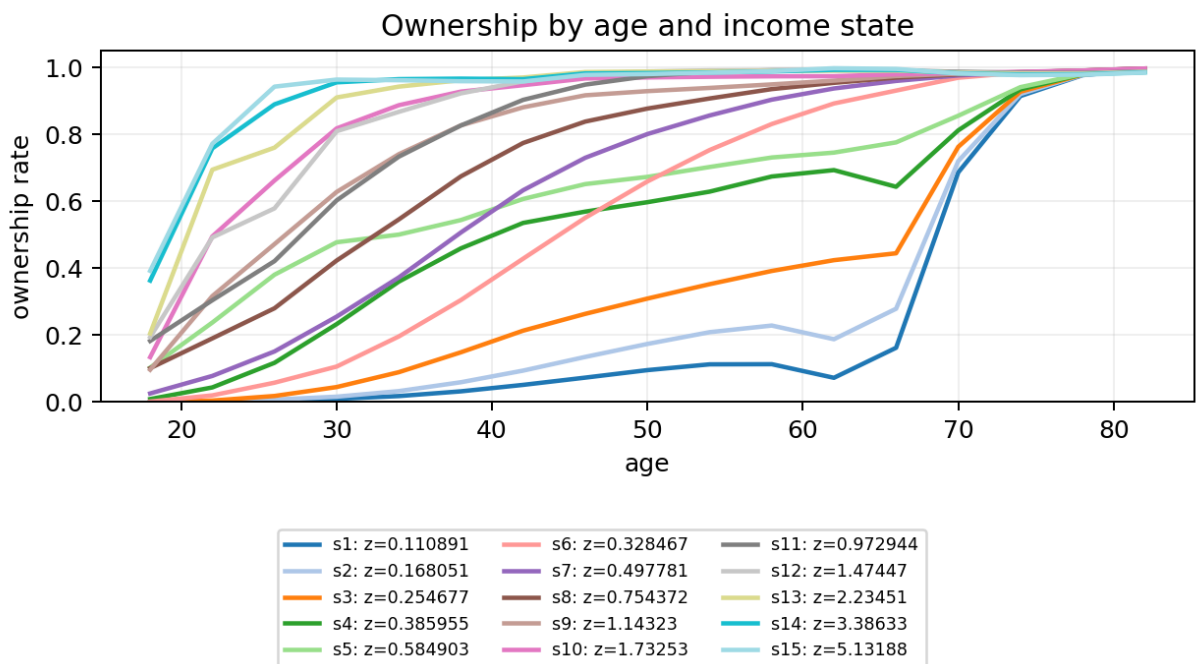
Annual 2% tax + equal rebate • 2063

Standard diagnostic 12: Ownership by age income state

Both policy paths at 2063. All other dates are in the companion graph set.



Annual 1% tax + equal rebate • 2063

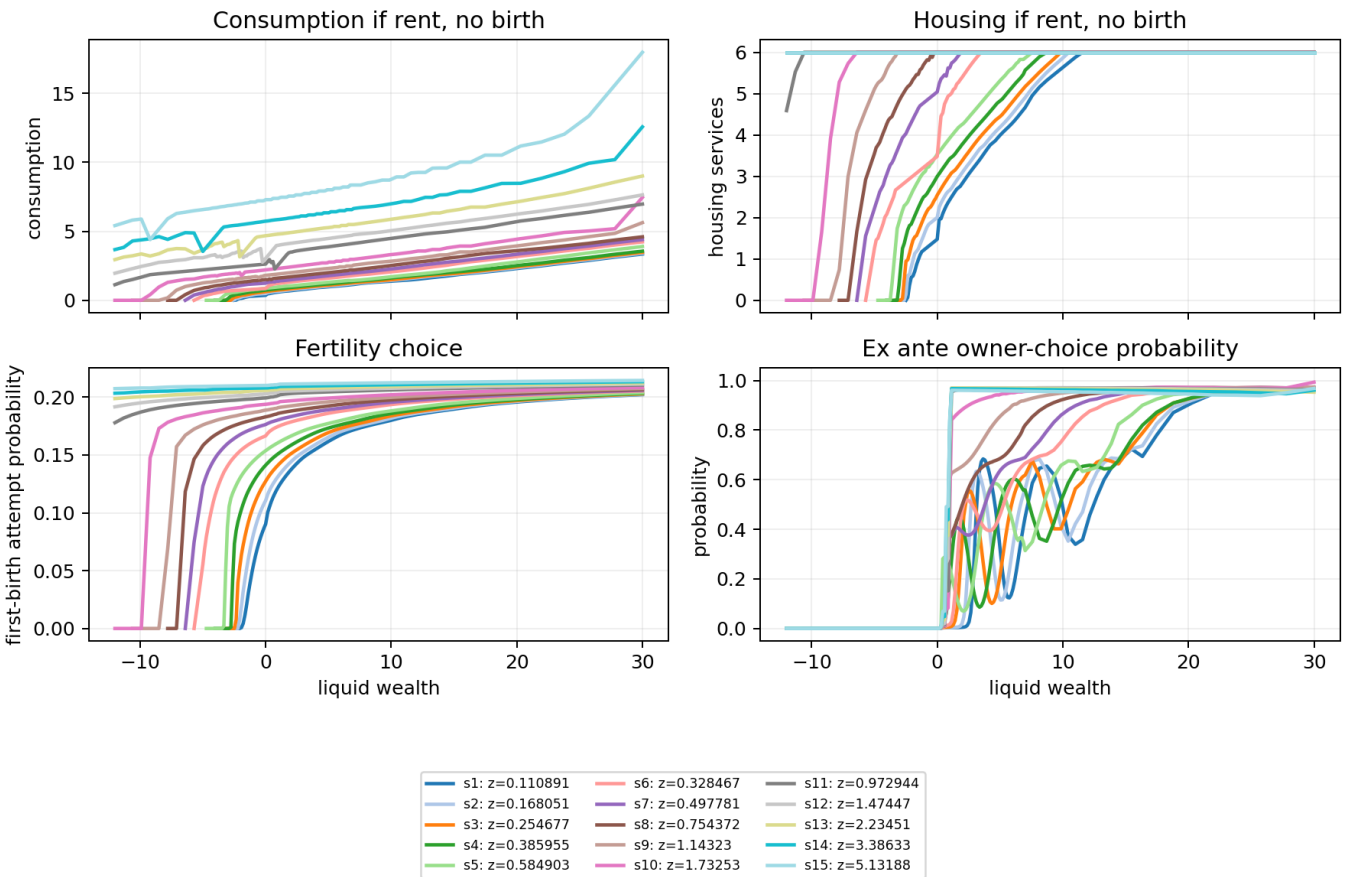


Annual 2% tax + equal rebate • 2063

Standard diagnostic 13: Policy childless renter age30

Both policy paths at 2063. All other dates are in the companion graph set. Consumption and housing are conditional on renting with no birth; fertility is the attempt probability. Ownership integrates over all housing products and conception outcomes. Policy panels include states with no households; boundary values require their conditional-feasibility interpretation.

Childless renter policies at age 30

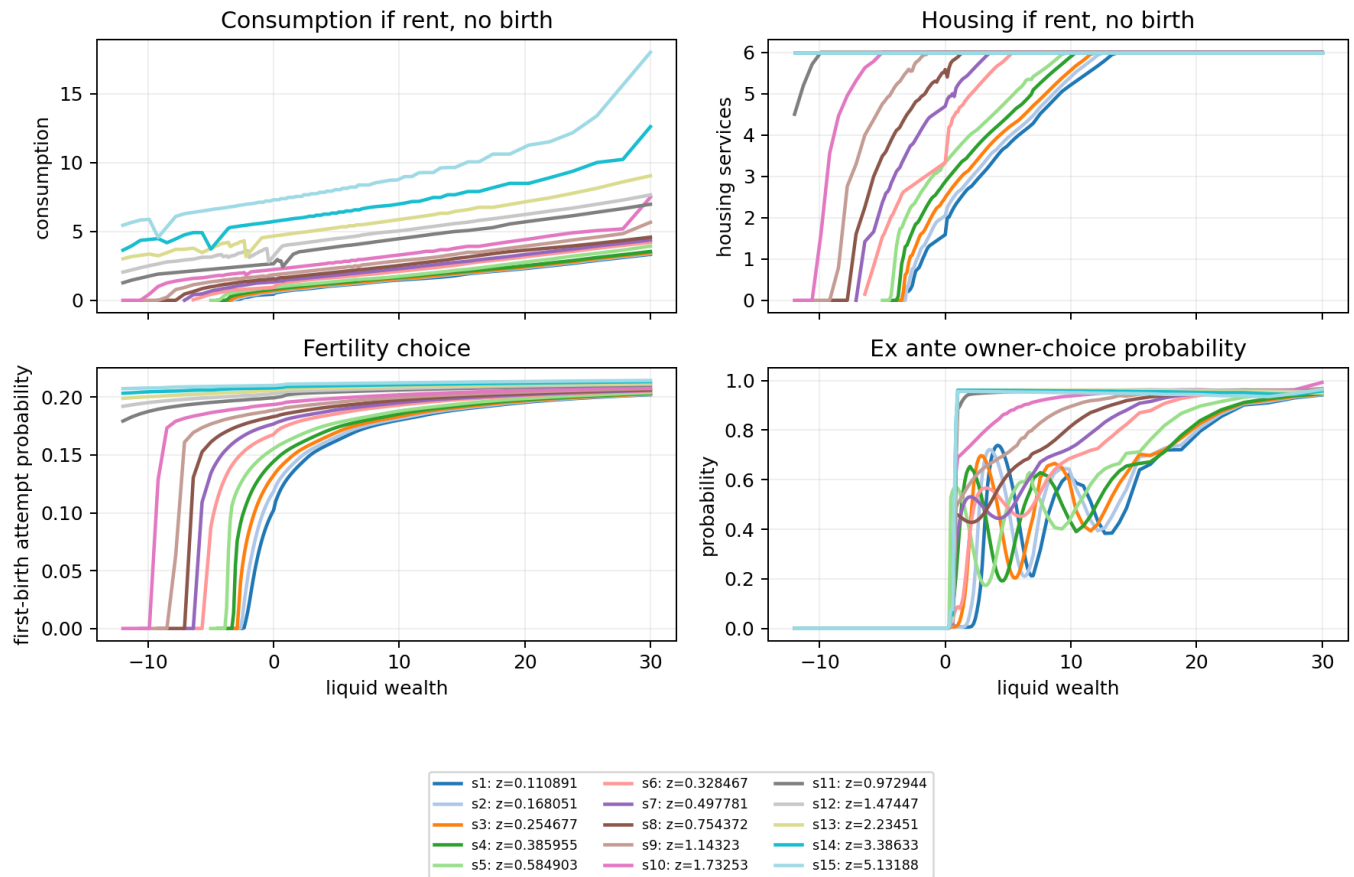


Annual 1% tax + equal rebate • 2063

Standard diagnostic 13: Policy childless renter age30 (continued)

Both policy paths at 2063. All other dates are in the companion graph set. Consumption and housing are conditional on renting with no birth; fertility is the attempt probability. Ownership integrates over all housing products and conception outcomes. Policy panels include states with no households; boundary values require their conditional-feasibility interpretation.

Childless renter policies at age 30

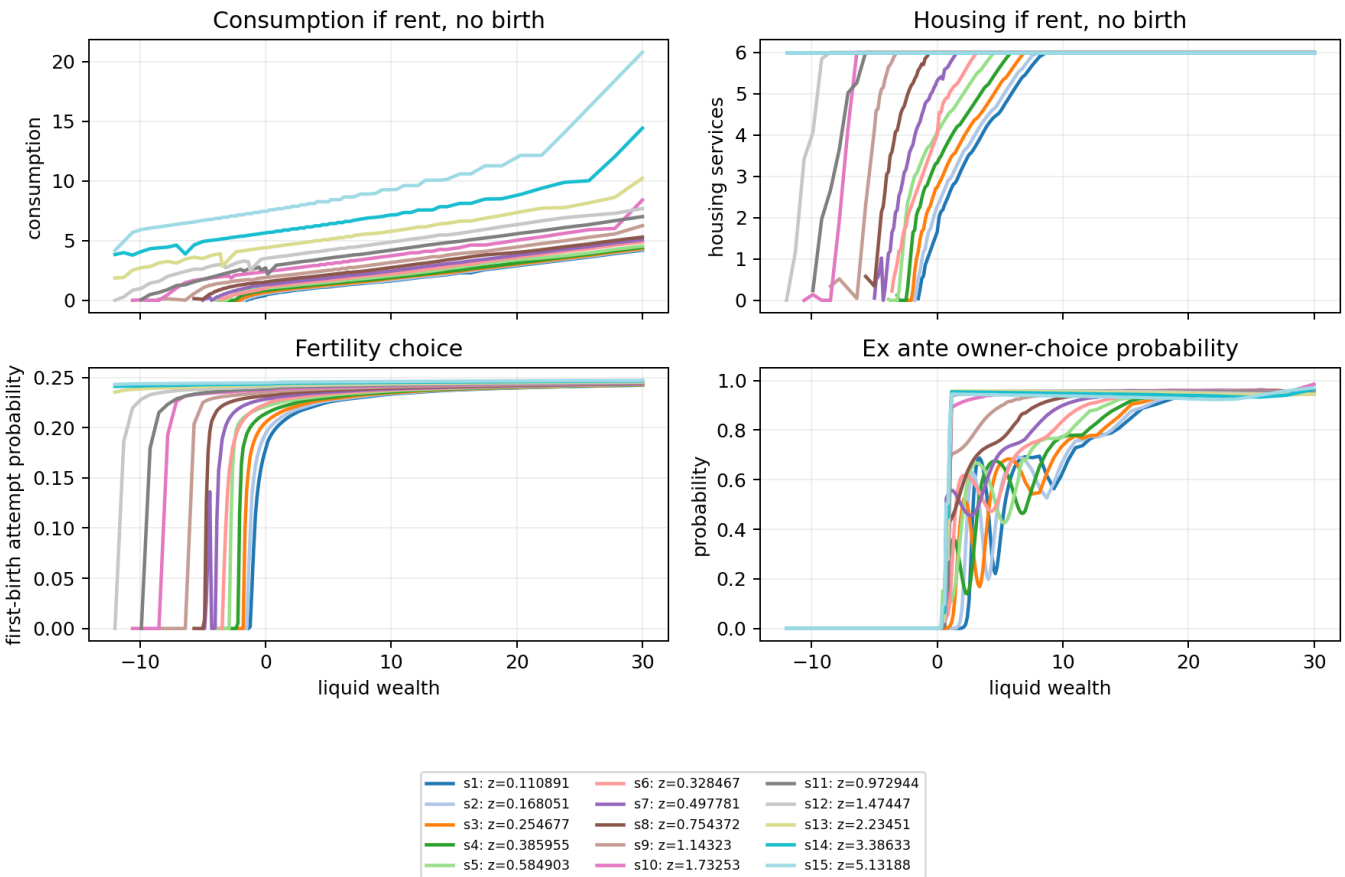


Annual 2% tax + equal rebate • 2063

Standard diagnostic 14: Policy childless renter age42

Both policy paths at 2063. All other dates are in the companion graph set. Consumption and housing are conditional on renting with no birth; fertility is the attempt probability. Ownership integrates over all housing products and conception outcomes. Policy panels include states with no households; boundary values require their conditional-feasibility interpretation.

Childless renter policies at age 42

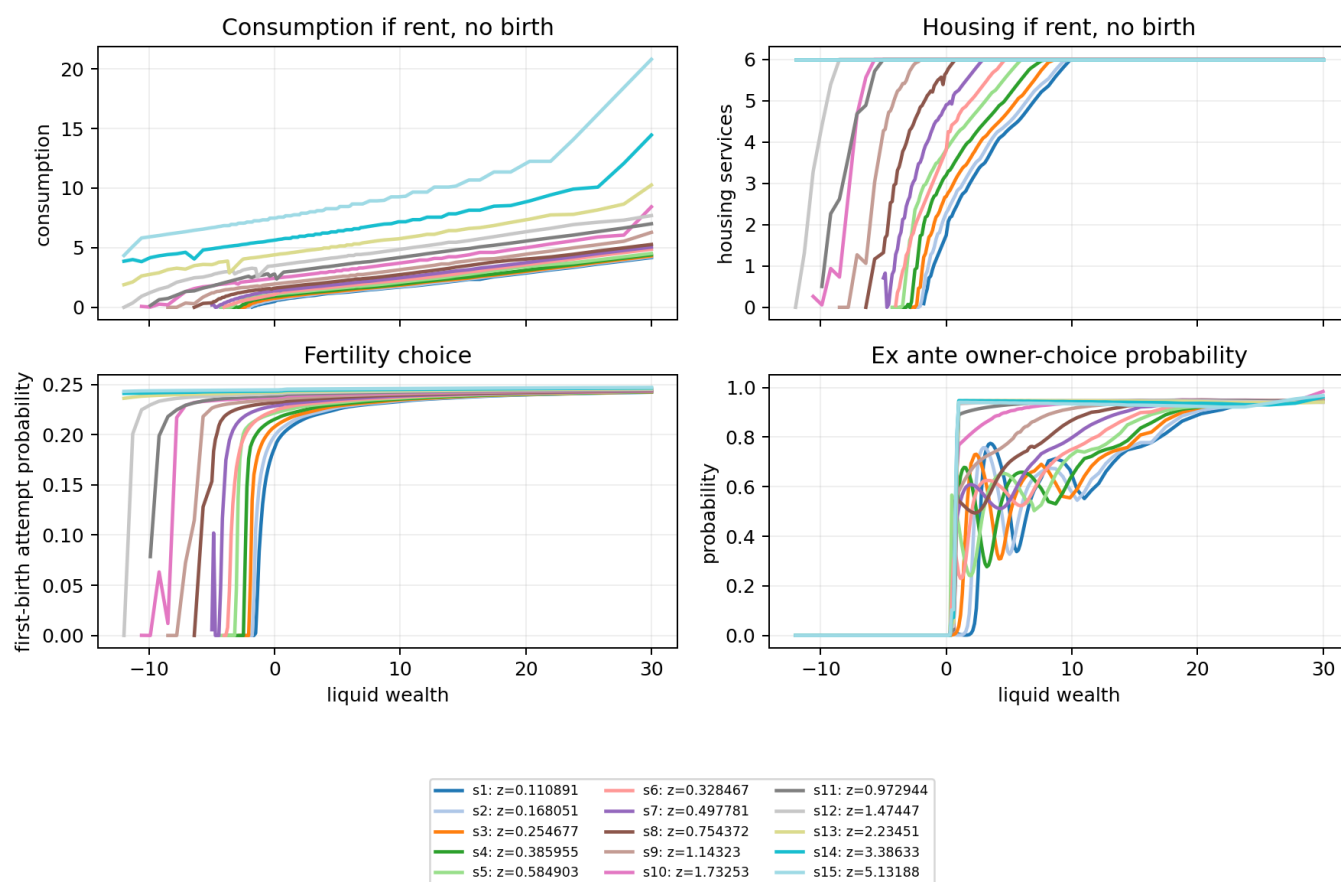


Annual 1% tax + equal rebate • 2063

Standard diagnostic 14: Policy childless renter age42 (continued)

Both policy paths at 2063. All other dates are in the companion graph set. Consumption and housing are conditional on renting with no birth; fertility is the attempt probability. Ownership integrates over all housing products and conception outcomes. Policy panels include states with no households; boundary values require their conditional-feasibility interpretation.

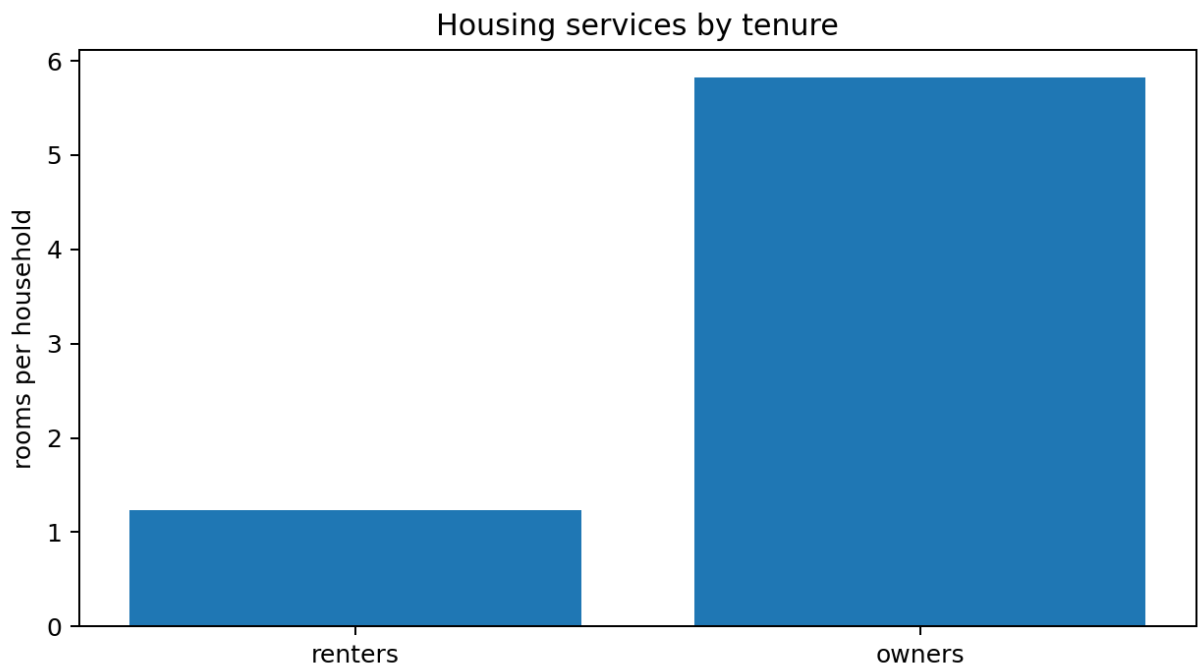
Childless renter policies at age 42



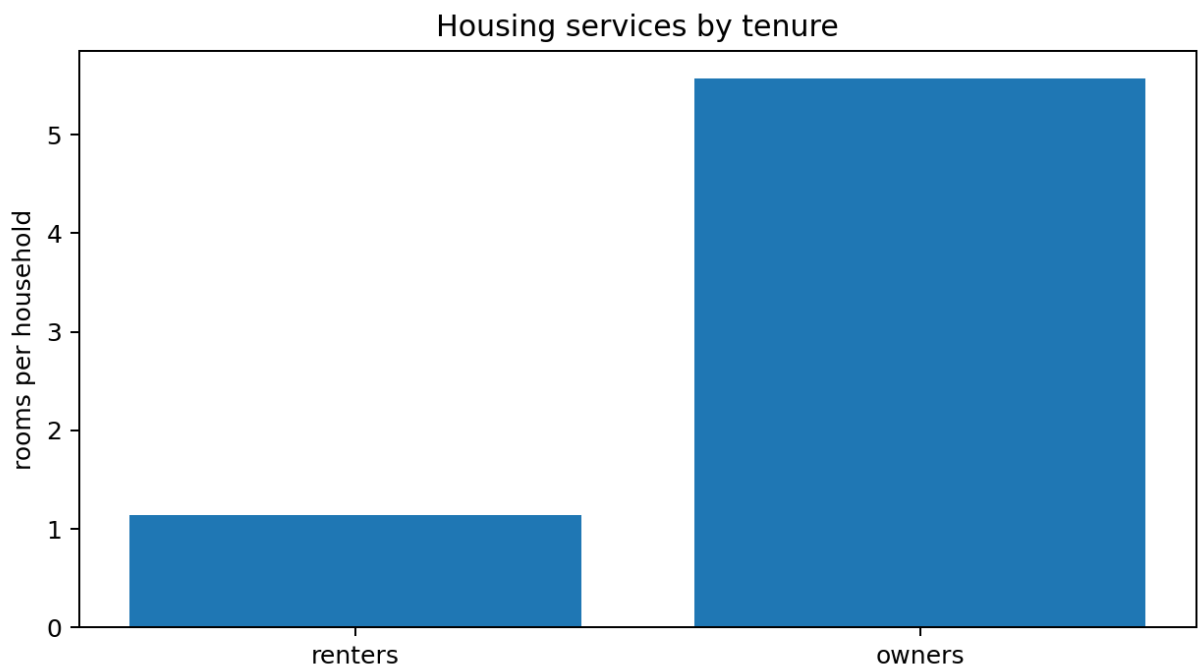
Annual 2% tax + equal rebate • 2063

Standard diagnostic 15: Tenure services

Both policy paths at 2063. All other dates are in the companion graph set. Bars are contributions to overall rooms per household, not conditional mean rooms within each tenure or housing size.



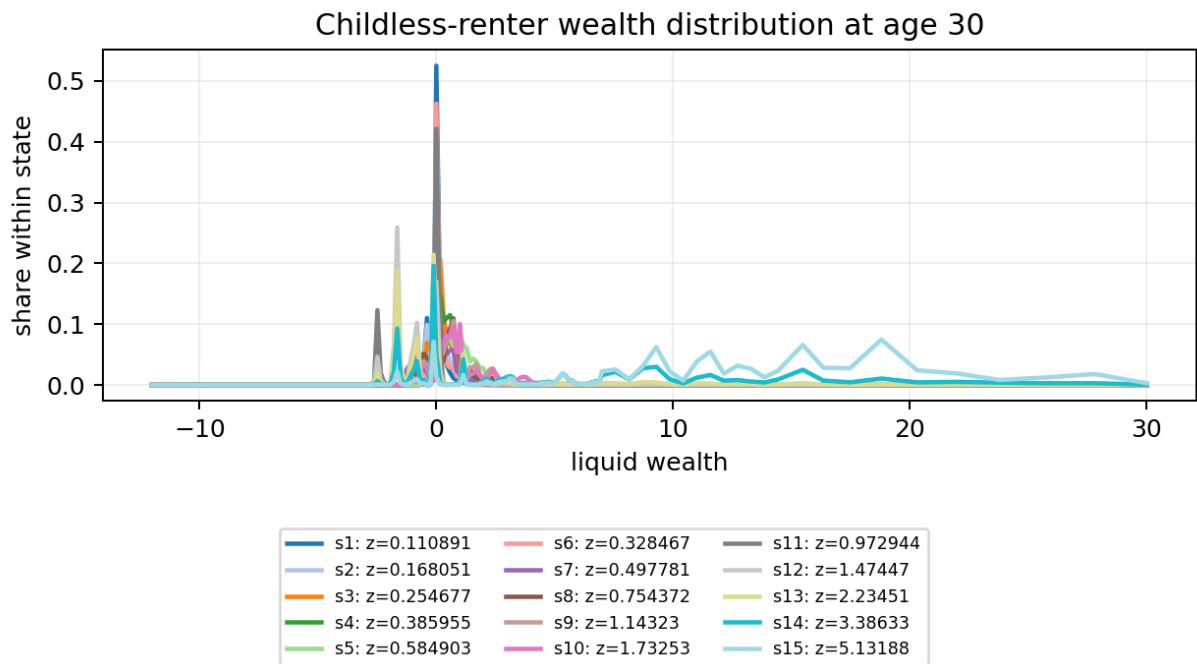
Annual 1% tax + equal rebate • 2063



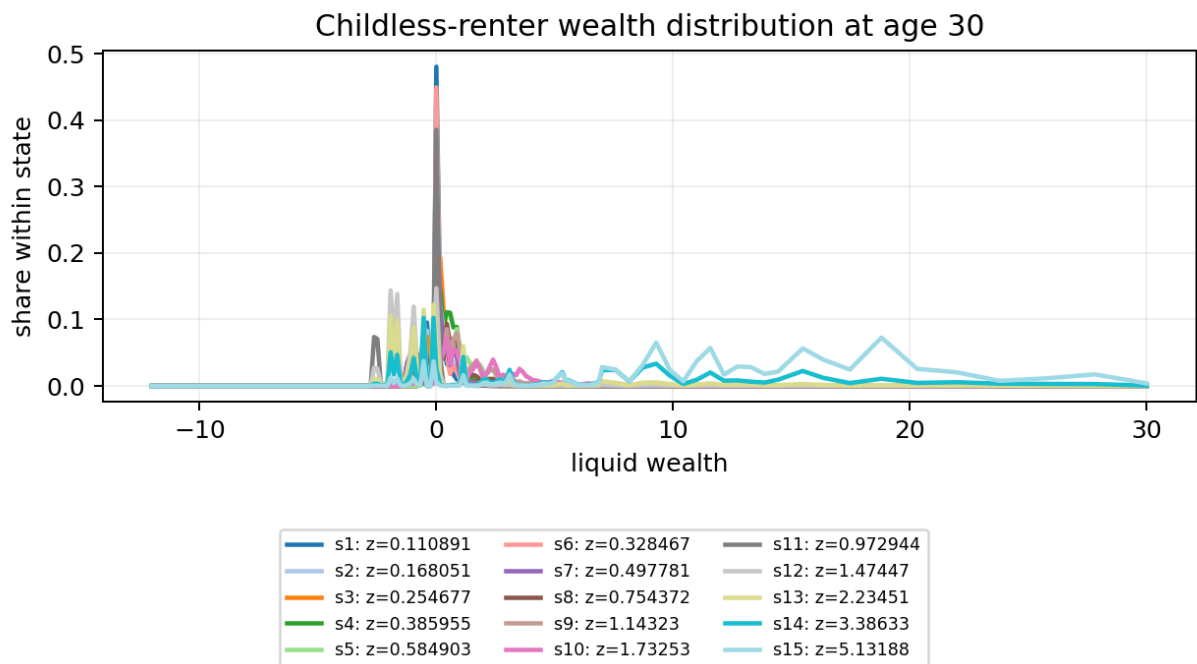
Annual 2% tax + equal rebate • 2063

Standard diagnostic 16: Wealth dist childless renter age30

Both policy paths at 2063. All other dates are in the companion graph set.



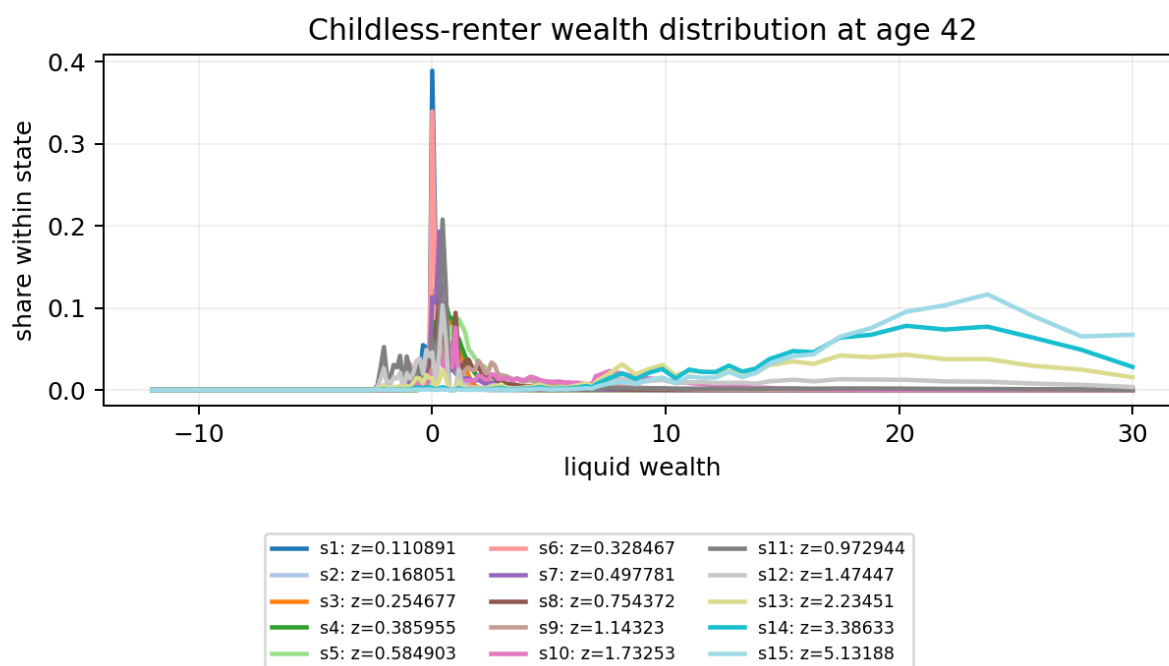
Annual 1% tax + equal rebate • 2063



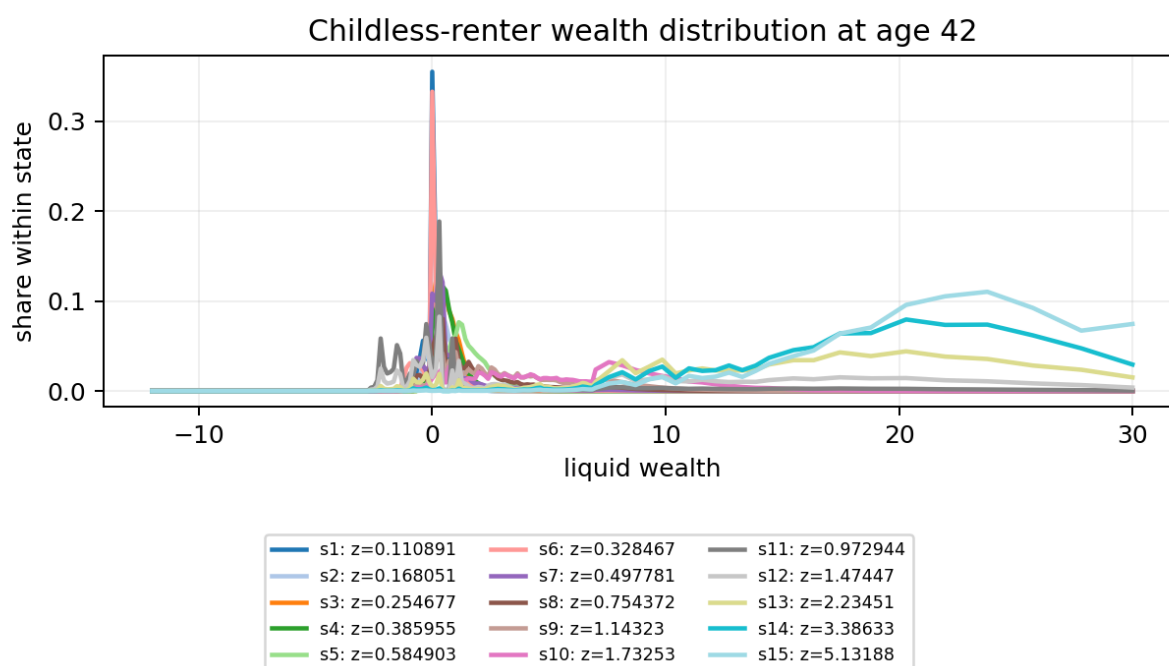
Annual 2% tax + equal rebate • 2063

Standard diagnostic 17: Wealth dist childless renter age42

Both policy paths at 2063. All other dates are in the companion graph set.



Annual 1% tax + equal rebate • 2063



Annual 2% tax + equal rebate • 2063